



# AUDIT REPORT

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December 2024

For



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# Executive Summary

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>          | WTTP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Project URL</b>           | <a href="https://linktr.ee/technicallyweb3">https://linktr.ee/technicallyweb3</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Overview</b>              | <p>Technically Web3 implements a data storage and publishing system on-chain that allows structured data to be stored, retrieved, and monetized through royalties. It defines data structures (DataPoint, DataPointStructure) that store metadata such as MIME type, charset, and storage location along with the actual data bytes. The DataPointStorage contract handles core storage operations including writing, reading, and retrieving metadata while using a hash-based addressing mechanism with collision handling.</p> <p>Additionally, the DataPointRegistry contract introduces an economic layer where publishers can register data points and earn royalties whenever their stored data is reused. The royalty amount is calculated based on the gas used to store the data and a predefined royalty rate. This design enables a decentralized data publishing system where contributors can be compensated for sharing data while ensuring persistent and verifiable on-chain storage references.</p> |
| <b>Audit Scope</b>           | <a href="https://github.com/TechnicallyWeb3/wttp/blob/patch_publisher-royalty-fix/contracts/WebStorage.sol">https://github.com/TechnicallyWeb3/wttp/blob/patch_publisher-royalty-fix/contracts/WebStorage.sol</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Contracts in Scope</b>    | WebStorage.sol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Commit Hash</b>           | <a href="#">693ed33a6bdd3e899af5f120ed2efe74945638f1</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>              | Solidity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Blockchain</b>            | Ethereum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Method</b>                | Manual Analysis, Functional Testing, Automated Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Review 1</b>              | 2nd - 6th December, 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Updated Code Received</b> | 12th December, 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Review 2</b>              | 12th December, 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



**Fixed In**

[https://github.com/TechnicallyWeb3/wttp/blob/merge\\_main\\_methods/contracts/WebStorage.sol](https://github.com/TechnicallyWeb3/wttp/blob/merge_main_methods/contracts/WebStorage.sol)

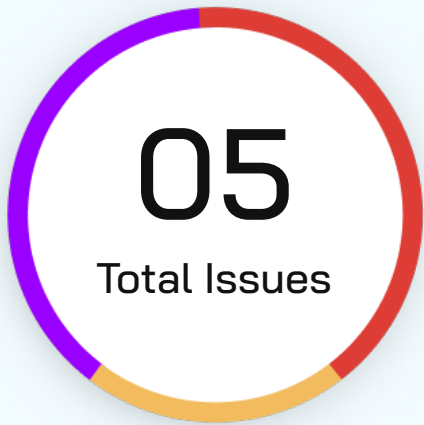
Commit: 8b8105942738af112c0c99a43f7b5179d6293c60

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# Number of Issues per Severity



|               |           |
|---------------|-----------|
| Critical      | 0 (0.0%)  |
| High          | 2 (40.0%) |
| Medium        | 0 (0.0%)  |
| Low           | 1 (20.0%) |
| Informational | 2 (40.0%) |

|        |                    | Severity |      |        |     |               |
|--------|--------------------|----------|------|--------|-----|---------------|
|        |                    | Critical | High | Medium | Low | Informational |
| Issues | Open               | 0        | 0    | 0      | 0   | 0             |
|        | Acknowledged       | 0        | 2    | 0      | 1   | 0             |
|        | Partially Resolved | 0        | 0    | 0      | 0   | 0             |
|        | Resolved           | 0        | 0    | 0      | 0   | 2             |



# Summary of Issues

| Issue No. | Issue Title                                                          | Severity      | Status       |
|-----------|----------------------------------------------------------------------|---------------|--------------|
| 1         | _royaltyGasRate() can be inaccurate across chains                    | High          | Acknowledged |
| 2         | Gas calculation vulnerabilities in royalty fee calculation mechanism | High          | Acknowledged |
| 3         | Cannot reset the publisher and royalty metrics                       | Low           | Acknowledged |
| 4         | Modularize code                                                      | Informational | Resolved     |
| 5         | _useFileSystem can be removed                                        | Informational | Resolved     |



# Checked Vulnerabilities

✓ Access Management

✓ Arbitrary write to storage

✓ Centralization of control

✓ Ether theft

✓ Improper or missing events

✓ Logical issues and flaws

✓ Arithmetic Computations  
Correctness

✓ Race conditions/front running

✓ SWC Registry

✓ Re-entrancy

✓ Timestamp Dependence

✓ Gas Limit and Loops

✓ Exception Disorder

✓ Gasless Send

✓ Use of tx.origin

✓ Malicious libraries

✓ Compiler version not fixed

✓ Address hardcoded

✓ Divide before multiply

✓ Integer overflow/underflow

✓ ERC's conformance

✓ Dangerous strict equalities

✓ Tautology or contradiction

✓ Return values of low-level calls



✓ Missing Zero Address Validation

✓ Private modifier

✓ Revert/require functions

✓ Multiple Sends

✓ Using suicide

✓ Using delegatecall

✓ Upgradeable safety

✓ Using throw

✓ Using inline assembly

✓ Style guide violation

✓ Unsafe type inference

✓ Implicit visibility level





# Techniques and Methods

**Throughout the audit of smart contracts, care was taken to ensure:**

- The overall quality of code
- Use of best practices
- Code documentation and comments, match logic and expected behavior
- Token distribution and calculations are as per the intended behavior mentioned in the whitepaper
- Efficient use of gas
- Code is safe from re-entrancy and other vulnerabilities

**The following techniques, methods, and tools were used to review all the smart contracts:**

## Structural Analysis

In this step, we have analyzed the design patterns and structure of smart contracts. A thorough check was done to ensure the smart contract is structured in a way that will not result in future problems.

## Static Analysis

A static Analysis of Smart Contracts was done to identify contract vulnerabilities. In this step, a series of automated tools are used to test the security of smart contracts.



### Code Review / Manual Analysis

Manual Analysis or review of code was done to identify new vulnerabilities or verify the vulnerabilities found during the static analysis. Contracts were completely manually analyzed, their logic was checked and compared with the one described in the whitepaper. Besides, the results of the automated analysis were manually verified.

### Gas Consumption

In this step, we have checked the behavior of smart contracts in production. Checks were done to know how much gas gets consumed and the possibilities of optimization of code to reduce gas consumption.

### Tools and Platforms Used for Audit

Remix IDE, Foundry, Solhint, Mythril, Slither, Solidity Static Analysis.



# Types of Severity

Every issue in this report has been assigned to a severity level. There are five levels of severity, and each of them has been explained below.

## ■ **Critical: Immediate and Catastrophic Impact**

Critical issues are the ones that an attacker could exploit with relative ease, potentially leading to an immediate and complete loss of user funds, a total takeover of the protocol's functionality, or other catastrophic failures. Critical vulnerabilities are non-negotiable; they absolutely must be fixed.

## ■ **High (H): Significant Risk of Major Loss or Compromise**

High-severity issues represent serious weaknesses that could result in significant financial losses for users, major malfunctions within the protocol, or substantial compromise of its intended operations. While exploiting these vulnerabilities might require specific conditions to be met or a moderate level of technical skill, the potential damage is considerable. These findings are critical and should be addressed and resolved thoroughly before the contract is put into the Mainnet.

## ■ **Medium (M): Potential for Moderate Harm Under Specific Circumstances**

Medium-severity bugs are loopholes in the protocol that could lead to moderate financial losses or partial disruptions of the protocol's intended behavior. However, exploiting these vulnerabilities typically requires more specific and less common conditions to occur, and the overall impact is generally lower compared to high or critical issues. While not as immediately threatening, it's still highly recommended to address these findings to enhance the contract's robustness and prevent potential problems down the line.

## ■ **Low (L): Minor Imperfections with Limited Repercussions**

Low-severity issues are essentially minor imperfections in the smart contract that have a limited impact on user funds or the core functionality of the protocol. Exploiting these would usually require very specific and unlikely scenarios and would yield minimal gain for an attacker. While these findings don't pose an immediate threat, addressing them when feasible can contribute to a more polished and well-maintained codebase.

## ■ **Informational (I): Opportunities for Improvement, Not Immediate Risks**

Informational findings aren't security vulnerabilities in the traditional sense. Instead, they highlight areas related to the clarity and efficiency of the code, gas optimization, the quality of documentation, or adherence to best development practices. These findings don't represent any immediate risk to the security or functionality of the contract but offer valuable insights for improving its overall quality and maintainability. Addressing these is optional but often beneficial for long-term health and clarity.



# Types of Issues

## Open

Security vulnerabilities identified that must be resolved and are currently unresolved.

## Resolved

These are the issues identified in the initial audit and have been successfully fixed.

## Acknowledged

Vulnerabilities which have been acknowledged but are yet to be resolved.

## Partially Resolved

Considerable efforts have been invested to reduce the risk/impact of the security issue, but are not completely resolved.

# Severity Matrix

|            |        | Impact   |        |        |
|------------|--------|----------|--------|--------|
|            |        | High     | Medium | Low    |
| Likelihood | High   | Critical | High   | Medium |
|            | Medium | High     | Medium | Low    |
|            | Low    | Medium   | Low    | Low    |

## Impact

- **High** - leads to a significant material loss of assets in the protocol or significantly harms a group of users.
- **Medium** - only a small amount of funds can be lost (such as leakage of value) or a core functionality of the protocol is affected.
- **Low** - can lead to any kind of unexpected behavior with some of the protocol's functionalities that's not so critical.

## Likelihood

- **High** - attack path is possible with reasonable assumptions that mimic on-chain conditions, and the cost of the attack is relatively low compared to the amount of funds that can be stolen or lost.
- **Medium** - only a conditionally incentivized attack vector, but still relatively likely.
- **Low** - has too many or too unlikely assumptions or requires a significant stake by the attacker with little or no incentive.



# High Severity Issues

## **`_royaltyGasRate()` can be inaccurate across chains**

**Acknowledged**

### Path

WebStorage#L125

### Function Name

`_royaltyGasRate()`

### Description

A static royalty gas rate introduces a vulnerability across chains that halts the protocol during its data retrieval phase. Gas costs on Ethereum are more significant than they are on chains like Polygon, Arbitrum and Binance Smart Chain.

```
function _royaltyGasRate() internal pure virtual returns (uint256) {  
    return 0.01 gwei  
}
```

The 0.01 gwei constant may appear safe for Ethereum but would become prohibitively expensive across other chains that have lesser gas fee requirements.

### Recommendation

Implement variable gas estimation systems using decentralized oracles to fetch EVM-chain gas fees, like Chainlink and BlockNative.

### WTPP Team's Comments

The Registry is a V0.1 and we acknowledge it can be updated in the future, we will aim to make DPR on the virtual web server contracts modifiable. It is important that these be immutable since these will be public contracts. Since it's a V0.1 we acknowledge that Chainlink will be better though there is a potential failure point (LINK balance) with it and so the V0.1 will be static, however a V0.2 will still use V0.1 for any legacy data but will extend the royalty data including rate as determined per chain 0.1-1% of average gas fees. the V0.2 will likely have DAO governance which will be able to adjust rate, protocol fee and override royalty addresses in the V0.1 contract.



## Gas calculation vulnerabilities in royalty fee calculation mechanism

**Acknowledged**

### Path

WebStorage#L129

### Function Name

`getRoyalty()`

### Description

Without an upper bound on gasUsed, if the value is manipulated to an arbitrarily high amount there would be greater costs incurred for fee retrieval. Users could be discouraged from utilizing the protocol to store and retrieve data and opt-out for cheaper larger scale alternatives. The static nature of the gas rate compounds the problem, especially across different blockchain networks with varying gas pricing models. What might be a reasonable calculation on one chain could become a critical vulnerability on another. The current implementation fails to account for the dynamic and fluctuating nature of gas prices across different EVM-compatible blockchains, creating inconsistent and potentially exploitable royalty mechanisms. Moreover, the current approach does not implement any safeguards against computational attacks.

There are no checks to prevent excessive gas usage, no mechanism to cap potentially malicious calculations, and no protection against integer arithmetic vulnerabilities. This leaves the contract exposed to sophisticated economic manipulation strategies that could compromise the intended royalty distribution mechanism.

### Recommendation

- Implement a max gas threshold relative to the amount of gas used.  
First, introduce a maximum gas threshold that caps the potentially exploitable gasUsed value. This prevents attackers from artificially inflating gas consumption beyond a reasonable limit. The threshold should be carefully calibrated based on expected use cases and typical computational requirements of the contract.
- Refer to issue H.1, compare and test the max limits for gas consumed and multiply by the royalty gas rate to have a holistic example of the costs incurred for royalties per chain.

Implementing a chain-aware pricing model/decentralized oracle fee retrieval mechanism that adjusts gas rates based on the specific blockchain network. By implementing a mapping of chain IDs to appropriate gas rates, the contract can maintain consistent and fair royalty calculations across various EVM-compatible chains.

### WTPP Team's Comments

Same comments in Issue 1.



# Low Severity Issues

## Cannot reset the publisher and royalty metrics

**Resolved**

### Path

WebStorage.sol#L93-L95

### Description

When the datapoint is written to for the first time the publisher and gas used amount are set (or left blank to be royalty free), but if such a feature is to be implemented for royalty free data retrievals for a portion of time and then royalties get added, this cannot be implemented because there is no setter available after the first write.

This would lead to the data having to be re-written to a new storage location (datapoint) and will need users to update the datapoint to read from the new storage location.

### Recommendation

Include a setter for the DataPointRoyalty struct.

### WTPP Team's Comments

The Registry is a V0.1 and we acknowledge it can be updated in the future, we will aim to make DPR on the virtual web server contracts modifiable. It is important that these be immutable since these will be public contracts. Since it's a V0.1 we acknowledge that Chainlink will be better though there is a potential failure point (LINK balance) with is and so the V0.1 will be static, however a V0.2 will still use V0.1 for any legacy data but will extend the royalty data including rate as determined per chain 0.1-1% of average gas fees. the V0.2 will likely have DAO governance which will be able to adjust rate, protocol fee and override royalty addresses in the V0.1 contract.





# Informational Issues

## Modularize code

**Resolved**

### Path

WebStorage.sol#L4

### Description

The DataPointInfo struct contains 4 data fields [size, mimeType, charset, location] of which 3 make up the DataPointStructure. This increases the number of lines of code unnecessarily, and reduces the value of having the DataPointStructure data structure created.

### Recommendation

DataPointInfo can use DataPointStructure within its struct for modularity and ease of access during function calls.

## \_useFileSystem can be removed

**Resolved**

### Path

WebStorage.sol#L113

### Function Name

\_useFileSystem

### Description

The internal function described as \_useFileSystem only returns the reference passed to it (a DataPointStorage) as the same value. It is only used within the writeDataPoint function to return the same DataPointStorage value already in storage. In the current implementation, it is redundant and can be removed if the internal function is not intended to be used anywhere else in the codebase.

### Recommendation

Remove the \_useFileSystem function.



# Automated Tests

No major issues were found. Some false positive errors were reported by the tools. All the other issues have been categorized above according to their level of severity.



# Closing Summary

In this report, we have considered the security of WTTP. We performed our audit according to the procedure described above.

Some issues of High,low and informational severity were found. Some suggestions, gas optimizations and best practices are also provided in order to improve the code quality and security posture. In the End, WTTP Team, Resolved two issues and Acknowledged the others.

# Disclaimer

At QuillAudits, we have spent years helping projects strengthen their smart contract security. However, security is not a one-time event—threats evolve, and so do attack vectors. Our audit provides a security assessment based on the best industry practices at the time of review, identifying known vulnerabilities in the received smart contract source code.

This report does not serve as a security guarantee, investment advice, or an endorsement of any platform. It reflects our findings based on the provided code at the time of analysis and may no longer be relevant after any modifications. The presence of an audit does not imply that the contract is free of vulnerabilities or fully secure.

While we have conducted a thorough review, security is an ongoing process. We strongly recommend multiple independent audits, continuous monitoring, and a public bug bounty program to enhance resilience against emerging threats.

Stay proactive. Stay secure.



# About QuillAudits

QuillAudits is a leading name in Web3 security, offering top-notch solutions to safeguard projects across DeFi, GameFi, NFT gaming, and all blockchain layers.

With eight years of expertise, we've secured over 1500 projects globally, averting over \$3 billion in losses. Our specialists rigorously audit smart contracts and ensure DApp safety on major platforms like Ethereum, BSC, Arbitrum, Algorand, Tron, Polygon, Polkadot, Fantom, NEAR, Solana, and others, guaranteeing your project's security with cutting-edge practices.

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December 2024

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